VLM-as-a-Judge Baselines

Design, Motivation, What Each Comparison Proves, and Results

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Abstract

This report documents every baseline and comparison policy implemented in the multimodal selective-evaluation experiments. The task is pairwise VLM judging: given an image, an instruction, and two candidate responses, a judge predicts the response preferred by human annotators. The benchmark compares four model families, four confidence-selection strategies, four prompt/order reliability setups, a consistency-escalation policy, the original stage-wise certified cascade, and two end-to-end certified routers. For each comparison, the report states the decision rule, why the baseline is necessary, and the hypothesis it tests. It also records the corrected 1,000-split four-judge results and identifies implemented policies whose post-fix results have not yet been generated.

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1 Experimental question and protocol

The central question is not only whether a VLM can agree with humans. It is whether a system can answer a useful fraction of comparisons while guaranteeing a user-selected level of human agreement. The experiment sets the maximum selective error to $\alpha = 0.15$, so the accuracy target is $1 - \alpha = 85\%$. The certificate uses $\delta = 0.10$, corresponding to a target confidence level of $1 - \delta = 90\%$.

The fast benchmark contains 1,000 approximately source-stratified, image-grounded preference pairs. Each scored example contains three simulated-annotator prompts, one few-shot demonstration per prompt, and both response orderings. The judge chain is:

Granite Vision 3.3 2B → Qwen2.5-VL 3B Instruct → Phi-3.5 Vision Instruct → InternVL3 8B

The chain deliberately spans different model families. This reduces the chance that a result is merely a within-family scaling artifact and lets disagreement between model families serve as a reliability signal. All policies operate on the same cached model probabilities. Re-running policy evaluation requires no VLM inference.

1.1 Metrics

Let A be the event that a policy answers and let $\widehat{Y}$ and Y denote the policy prediction and human label. The primary selective metric is

$$\text{Accuracy} = \Pr(\widehat{Y} = Y \mid A),$$

which is undefined for an empty selection and is therefore averaged only over nonempty test splits. Coverage is $\Pr(A)$. Macro accuracy first computes accuracy within each data source and then gives every source equal weight. Nonempty rate is the fraction of repeated splits that answer at least one example. Target success is the fraction of all splits that both answer and attain at least 85% test accuracy. For heuristic and point-estimate methods, "success" is empirical and is not a formal certificate.

2 Why this baseline suite is necessary

A single accuracy comparison cannot locate the source of improvement. The suite is organized around five distinct questions:

Axis	Required comparison	What it identifies
Judge strength	Direct use of every model	Whether gains come from a stronger judge rather than selection or routing.
Confidence selection	Heuristic, point estimate, and certified selectors	Whether confidence ranks correctness, and what finite-sample certification costs in coverage.
Cascading	Single-judge selectors versus sequential policies	Whether multiple model families add coverage beyond the strongest judge and what escalation costs.
Prompt reliability	Single prompt, order averaging, voting, and consistency filtering	Whether prompt variation and position bias explain confidence and accuracy.
Certificate scope	Stage-wise cascade versus global routers	Whether coverage is lost by protecting several stage claims instead of the deployed policy's final risk.

The direct pairwise protocol follows multimodal judge and reward-model benchmarks that measure agreement with human preferences [2, 3, 4]. The selective and cascaded comparisons follow *Trust or Escalate* [1]. Prompt/order ablations operationalize reliability mechanisms used in multimodal reward benchmarking and repeated judging. They are labeled as operational analogues when the cached deterministic runs do not exactly reproduce a paper's original stochastic protocol.

3 Model-strength baselines

3.1 Direct judge: `direct:<judge>`

Each of the four direct baselines chooses $\arg\max_y p_j(y \mid x)$ and never abstains. There is one instance for each model family; Appendix A lists every exact identifier.

Motivation. Pairwise accuracy at 100% coverage is the standard VLM-as-a-judge comparison. It establishes the raw ability of each family and provides the denominator against which abstention gains are judged.

What it proves. If a selective policy improves conditional accuracy, the direct rows show how much is attributable to abstaining. Comparing the four direct rows also tests whether the final judge is genuinely stronger. Direct baselines cannot demonstrate risk control because they make no coverage-dependent decision and use no certificate.

4 Confidence and selective-risk baselines

4.1 Strongest-judge heuristic: `heuristic:internvl3-8b`

The strongest judge answers when its cached confidence is at least $1 - \alpha = 0.85$.

Motivation. This is the simplest deployment rule a practitioner might use if model probabilities were calibrated. It has no learned threshold and no statistical bound.

What it proves. It tests whether confidence has useful ranking power and gives an empirical upper reference for coverage. A high success rate does not turn it into a certificate; it may fail under another calibration draw or distribution.

4.2 Cascaded heuristic: `cascaded_heuristic`

The same fixed threshold, 0.85, is applied sequentially. The first judge above the threshold answers; otherwise the item is escalated.

Motivation. This is the non-certified counterpart of a cascade. It tests whether sequential access to several families adds coverage beyond the strongest judge.

What it proves. Its difference from the strongest-judge heuristic isolates the empirical value of escalation. Its difference from the certified cascade quantifies the coverage price of finite-sample guarantees.

4.3 Point-estimate calibration: `point_estimate:<judge>`

For each judge, the method chooses the least restrictive threshold whose empirical calibration error is at most α, requiring at least 30 selected calibration examples. It does not add an upper confidence bound. Four instances are evaluated, one for each model family.

Motivation. This is a calibrated but non-certified comparator from the selective evaluation design. It separates threshold learning from statistical uncertainty.

What it proves. The gap between point-estimate and certified selection measures the coverage cost of accounting for finite calibration size. Failure on held-out splits shows why empirical calibration error alone is insufficient for a guarantee.

4.4 Certified single judge: `selective:<judge>`

Each judge independently uses fixed-sequence testing and the full $\delta = 0.1$ to find a threshold whose one-sided exact binomial upper risk bound is at most α. There is one instance for each model family; Appendix A lists every exact identifier.

Motivation. A single certified judge is the essential control for the cascade. It receives the full failure budget and does not benefit from other models.

What it proves. It separates gains from cascading from losses caused by splitting δ across stages. These policies are implemented in the current code, but they were added after the captured result file was generated.

Their rows are therefore marked "N/R" rather than filled with inferred values. Re-running CPU evaluation will produce them from the existing caches.

4.5 Original certified cascade: `cascaded_selective`

At stage j, fixed-sequence testing certifies a confidence threshold on the residual examples rejected by earlier stages. With four judges, the implementation assigns $\delta_j = \delta/4 = 0.025$ to each stage. If every stage-specific risk is at most α, the final risk is at most α with probability at least $1 - \delta$ by the union bound.

Motivation. This is the original proposed method and the principal comparator for the new global certificate [1].

What it proves. It tests whether a formally protected cascade can meet the human agreement target. Its coverage reveals the conservatism of simultaneously certifying multiple stages on shrinking residual sets.

5 Prompt and response-order reliability baselines

All four setup policies are instantiated separately for Granite, Qwen, Phi, and InternVL. They are post-hoc transformations of cached per-prompt, per-order probabilities, so their comparison requires no new model calls.

5.1 Vanilla single prompt: `setup:vanilla_single:<judge>`

Use the first simulated-annotator prompt and the original response order.

Motivation. This reproduces the common fixed-prompt judge setup used in basic pairwise evaluation [2, 3].

What it proves. It establishes how much the full prompt/order ensemble gains over one ordinary judge call. It also exposes position sensitivity because it never reverses the candidates.

5.2 Response-order averaging: `setup:position_swap:<judge>`

Evaluate the first prompt in both response orders, map both outputs back to the original candidate identities, average their probabilities, and always answer.

Motivation. Pairwise judges can prefer the first or second slot. Forward/reverse evaluation is used in multimodal reward benchmarking to measure and reduce this bias [5].

What it proves. Improvement over the vanilla row measures the benefit of explicit position control. Because it retains full coverage, it does not confound the comparison with abstention.

5.3 Prompt/order vote: `setup:prompt_order_vote:<judge>`

Take a majority vote across every cached simulated-annotator prompt and both response orders.

Motivation. Repeated judgments are a standard way to reduce prompt variance; VLRewardBench studies inference-time scaling for generative reward models [4].

What it proves. It tests whether diversity across prompts and response orders provides a robust ensemble signal. This implementation is a deterministic analogue over cached runs, not a reproduction of a stochastic five-call protocol.

5.4 Swap consistency: `setup:swap_consistency:<judge>`

Answer only when the first prompt selects the same original response in both response orders; otherwise abstain. For consistent pairs, average the two probability vectors.

Motivation. Self-consistency can serve as a label-free uncertainty signal and directly targets position-instability failures.

What it proves. Its accuracy/coverage curve tests whether order agreement identifies reliable examples. Comparing it with position averaging isolates the effect of filtering inconsistent items rather than merely averaging them.

5.5 Consistency escalation: `setup:consistency_escalation`

At each non-final stage, accept only swap-consistent judgments and route inconsistent examples onward. The final judge resolves every remaining pair using its full cached ensemble.

Motivation. Selective test-time evaluation allocates extra computation to self-inconsistent cases [8]. This policy tests that reusable routing idea without implementing an evolving meta-prompt.

What it proves. It tests whether consistency alone is sufficient for economical escalation. It is an operational analogue of the routing heuristic, not a full implementation of Learning While Evaluating.

6 End-to-end certified routing

6.1 Why change the scope of certification?

The stage-wise guarantee is valid, but it controls a stronger set of events than the deployed system requires. It separately requires

$$R_j = \Pr(\widehat{Y}_j \neq Y \mid \text{reached and accepted at stage } j) \leq \alpha$$

for every stage. The union bound protects all M claims with $\sum_j \delta_j = \delta$. This is conservative because each test uses a smaller δ_j, later stages receive fewer calibration examples, and every stage must meet α even though final risk is a coverage-weighted mixture of stage errors.

The global experiment freezes the complete policy π_τ and directly certifies

$$R(\pi_\tau) = \Pr(\widehat{Y}_{\pi_\tau} \neq Y \mid \pi_\tau \text{ answers}) \leq \alpha.$$

One policy-level claim can use the full δ. To prevent label leakage, every repetition uses disjoint group-safe router-training, certification, and test partitions. Threshold candidates are fixed from router-training scores before certification labels are examined.

6.2 Global raw confidence: `global_raw_confidence`

Route using each stage's cached maximum class probability, freeze a candidate threshold grid from the router-training split, and certify the final routing policy once.

Motivation. This is the minimal end-to-end modification. It changes certificate scope while keeping the original confidence signal unchanged.

What it proves. Its comparison with `cascaded_selective` isolates the benefit of a policy-level certificate. Its comparison with the learned router isolates whether richer reliability features help beyond raw confidence.

6.3 Global learned router: `global_learned_router`

Fit one regularized logistic correctness model per stage on the independent router split. Features are confidence, margin, entropy, simulated-annotator vote support, probability dispersion, response-order consistency and probability shift, agreement/unanimity among models visited so far, and previous-confidence summaries. The learned score is then frozen and certified exactly like the raw score.

Motivation. VLM reliability may be expressed by disagreement and response-order instability even when maximum probability is poorly calibrated.

What it proves. It tests whether cached multimodal reliability signals can improve routing without new inference. A loss against raw confidence is informative evidence that the 11-feature model has too much variance for the approximately 200-example router split.

7 Complete four-judge results

Table 2 reproduces the corrected post-fix baseline output. Accuracy excludes empty selections. The four certified single-judge policies are present in the current experiment registry but absent from the captured run, so no numbers are imputed.

Table 2: Four-judge baseline results over 1,000 repeated group-safe splits. Values are fractions. N/R means implemented but not reported in the captured run.

Policy	Acc.	Macro	Cov.	Nonempty	Success
direct:granite-vision-3.3-2b	.5909	.5971	1.0000	1.0000	.0000
direct:qwen2.5-vl-3b-instruct	.5762	.5988	1.0000	1.0000	.0000
direct:phi-3.5-vision-instruct	.5824	.6115	1.0000	1.0000	.0000
direct:internvl3-8b	.7051	.7325	1.0000	1.0000	.0000
heuristic:internvl3-8b	.8872	.8573	.3380	1.0000	.9990
cascaded_heuristic	.8695	.8477	.3611	1.0000	.9660
point_estimate:granite-vision-3.3-2b	.6816	.6931	.0099	.0580	.0000
point_estimate:qwen2.5-vl-3b-instruct	.8347	.7792	.1320	.8650	.3250
point_estimate:phi-3.5-vision-instruct	.7264	.7102	.0240	.1770	.0000
point_estimate:internvl3-8b	.8477	.8368	.4274	1.0000	.4570
selective:granite-vision-3.3-2b	N/R	N/R	N/R	N/R	N/R
selective:qwen2.5-vl-3b-instruct	N/R	N/R	N/R	N/R	N/R
selective:phi-3.5-vision-instruct	N/R	N/R	N/R	N/R	N/R
selective:internvl3-8b	N/R	N/R	N/R	N/R	N/R
cascaded_selective	.9362	.8793	.1306	.6680	.6370
setup:vanilla_single:granite-vision-3.3-2b	.4998	.5016	1.0000	1.0000	.0000
setup:position_swap:granite-vision-3.3-2b	.6125	.6394	1.0000	1.0000	.0000
setup:prompt_order_vote:granite-vision-3.3-2b	.5784	.5953	1.0000	1.0000	.0000
setup:swap_consistency:granite-vision-3.3-2b	.7640	.8096	.0681	1.0000	.0080
setup:vanilla_single:qwen2.5-vl-3b-instruct	.5830	.6045	1.0000	1.0000	.0000
setup:position_swap:qwen2.5-vl-3b-instruct	.5812	.6060	1.0000	1.0000	.0000
setup:prompt_order_vote:qwen2.5-vl-3b-instruct	.5883	.6072	1.0000	1.0000	.0000
setup:swap_consistency:qwen2.5-vl-3b-instruct	.6606	.6225	.4325	1.0000	.0000
setup:vanilla_single:phi-3.5-vision-instruct	.5641	.5899	1.0000	1.0000	.0000
setup:position_swap:phi-3.5-vision-instruct	.5709	.6025	1.0000	1.0000	.0000
setup:prompt_order_vote:phi-3.5-vision-instruct	.5772	.6012	1.0000	1.0000	.0000
setup:swap_consistency:phi-3.5-vision-instruct	.6379	.6628	.4940	1.0000	.0000
setup:vanilla_single:internvl3-8b	.6631	.6949	1.0000	1.0000	.0000
setup:position_swap:internvl3-8b	.7030	.7289	1.0000	1.0000	.0000
setup:prompt_order_vote:internvl3-8b	.6960	.7230	1.0000	1.0000	.0000
setup:swap_consistency:internvl3-8b	.7729	.7830	.6522	1.0000	.0000
setup:consistency_escalation	.6150	.6430	1.0000	1.0000	.0000

The two global policies use a three-way split (approximately 200 router-training, 300 certification, and 500 test examples), whereas the baseline table uses approximately 300 calibration and 700 test examples. The global policies are paired with each other on the same splits; comparisons with Table 2 are repeated-split aggregate comparisons rather than paired test-set differences.

Table 3: End-to-end certified routing over 1,000 repeated splits.

Method	Acc.	Macro	Cov.	Calls	Nonempty	Certified	Success
global_raw_confidence	.9108	.8658	.2520	3.891	.9600	.9600	.8650
global_learned_router	.9042	.8604	.1543	3.909	.6180	.6180	.5370

8 What the results establish

1. **Judge strength matters, but is insufficient.** InternVL is the strongest direct model at 70.51%, still well below the 85% target. A stronger final model alone does not solve reliable judging.
2. **Confidence is useful.** The InternVL heuristic reaches 88.72% accuracy at 33.80% coverage. The point-estimate version reaches greater coverage (42.74%) but only 84.77% mean accuracy and succeeds on 45.70% of splits. This is direct evidence that uncertainty in threshold calibration matters.
3. **Position control helps unevenly.** Granite improves from 49.98% with one prompt/order to 61.25% after order averaging. InternVL improves from 66.31% to 70.30%. Qwen and Phi change little. Position bias is therefore model dependent.
4. **Consistency is selective but not sufficient.** Swap consistency raises conditional accuracy for every judge, but only Granite attains a small nonzero target success rate. The full-coverage consistency-escalation analogue reaches 61.50%, showing that consistency alone is a weak routing signal in this chain.

5. **Stage-wise certification is accurate but conservative.** The original cascade reaches 93.62% accuracy, 8.62 points above target, at 13.06% coverage and a 66.80% nonempty rate. This large unused accuracy margin motivates a less conservative policy-level certificate.
6. **Global certification nearly doubles coverage.** Raw-confidence global certification raises coverage from 13.06% to 25.20% (1.93×) while maintaining 91.08% accuracy. Among certified splits, target success is $0.865/0.960 = 90.1\%$, matching the intended 90% confidence level empirically.
7. **The rich router is a negative ablation.** The learned router has lower coverage and certifies fewer splits than raw confidence. With only about 200 router examples, the 11-feature correctness model likely has too much variance.
8. **Compute efficiency remains unresolved.** The raw global policy calls 3.891 of four judges on average. Its main benefit is coverage from certificate scope, not early exit. Future routing work should report judge calls alongside accuracy and coverage.

9 Completeness and scope boundary

The experiment registry contains 34 distinct policy instances:

Policy family	**Instances**
Direct judges	4
Strongest-judge heuristic	1
Cascaded heuristic	1
Point-estimate single judges	4
Certified single judges	4
Original certified cascade	1
Vanilla single-prompt setups	4
Position-swap setups	4
Prompt/order-vote setups	4
Swap-consistency setups	4
Consistency escalation	1
Global raw-confidence router	1
Global learned router	1
Total	**34**

Prometheus-Vision [6] and LLaVA-Critic [7] are relevant specialist evaluator models, but they are not rows in the current cached experiment. Prometheus-Vision is trained for rubric-conditioned pointwise scoring, whose primary metric is correlation; adapting it to pairwise preference would change the protocol. LLaVA-Critic supports pairwise ranking, but a faithful comparison requires its official generative prompt/parser and new cached inference. Neither is silently presented as an implemented baseline. They should be added as separate model baselines only after their native protocols are validated on the same examples.

Multimodal RewardBench, MMRB2, MLLM-as-a-Judge, and VLRewardBench motivate the pairwise metrics and prompt/order controls; they are benchmark or setup references rather than additional model rows. Learning While Evaluating motivates the consistency-escalation heuristic; the current baseline does not claim its evolving meta-prompt mechanism.

10 Reproduction and reporting checklist

The complete cached-only evaluation command is:

```
python -u -m open_cascade evaluate configs/vlm_v2_fast_4judge.yaml
```

For a paper table:

- rerun the current code to populate the four `selective:<judge>` rows;
- retain 1,000 group-safe repetitions and report mean plus standard deviation;

- report accuracy, macro accuracy, coverage, macro coverage, nonempty rate, and target success;
- for global routing, also report certified rate, mean judge calls, early-exit rate, and model composition;
- state that raw/learned global comparisons are paired, while the original/global tables use different test sizes;
- label heuristic and point-estimate success as empirical rather than guaranteed;
- keep empty selections out of conditional-accuracy averages.

A Exact implemented policy identifiers

Family	Identifiers
Direct	`direct:granite-vision-3.3-2b`; `direct:qwen2.5-vl-3b-instruct`; `direct:phi-3.5-vision-instruct`; `direct:internvl3-8b`.
Heuristic	`heuristic:internvl3-8b`; `cascaded_heuristic`.
Point estimate	`point_estimate:granite-vision-3.3-2b`; `point_estimate:qwen2.5-vl-3b-instruct`; `point_estimate:phi-3.5-vision-instruct`; `point_estimate:internvl3-8b`.
Certified single judge	`selective:granite-vision-3.3-2b`; `selective:qwen2.5-vl-3b-instruct`; `selective:phi-3.5-vision-instruct`; `selective:internvl3-8b`.
Certified cascade	`cascaded_selective`.
Vanilla setup	`setup:vanilla_single:granite-vision-3.3-2b`; `setup:vanilla_single:qwen2.5-vl-3b-instruct`; `setup:vanilla_single:phi-3.5-vision-instruct`; `setup:vanilla_single:internvl3-8b`.
Position swap	`setup:position_swap:granite-vision-3.3-2b`; `setup:position_swap:qwen2.5-vl-3b-instruct`; `setup:position_swap:phi-3.5-vision-instruct`; `setup:position_swap:internvl3-8b`.
Prompt/order vote	`setup:prompt_order_vote:granite-vision-3.3-2b`; `setup:prompt_order_vote:qwen2.5-vl-3b-instruct`; `setup:prompt_order_vote:phi-3.5-vision-instruct`; `setup:prompt_order_vote:internvl3-8b`.
Swap consistency	`setup:swap_consistency:granite-vision-3.3-2b`; `setup:swap_consistency:qwen2.5-vl-3b-instruct`; `setup:swap_consistency:phi-3.5-vision-instruct`; `setup:swap_consistency:internvl3-8b`.
Consistency escalation	`setup:consistency_escalation`.
Global routers	`global_raw_confidence`; `global_learned_router`.

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